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Drishti - Assistive Navigation System for Visually Impaired People

Theme: IoT and Sensors | Project No.: 63 | Experiential Learning Phase 1

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PROJECT TEAM

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PROJECT TIMELINE

Presentation Outline

1. Introduction
2. Literature Review
3. Problem Definition
4. Objectives
5. Methodology
6. Proposed Outcomes
7. Timeline of the project
8. Conclusion
9. References

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Introduction

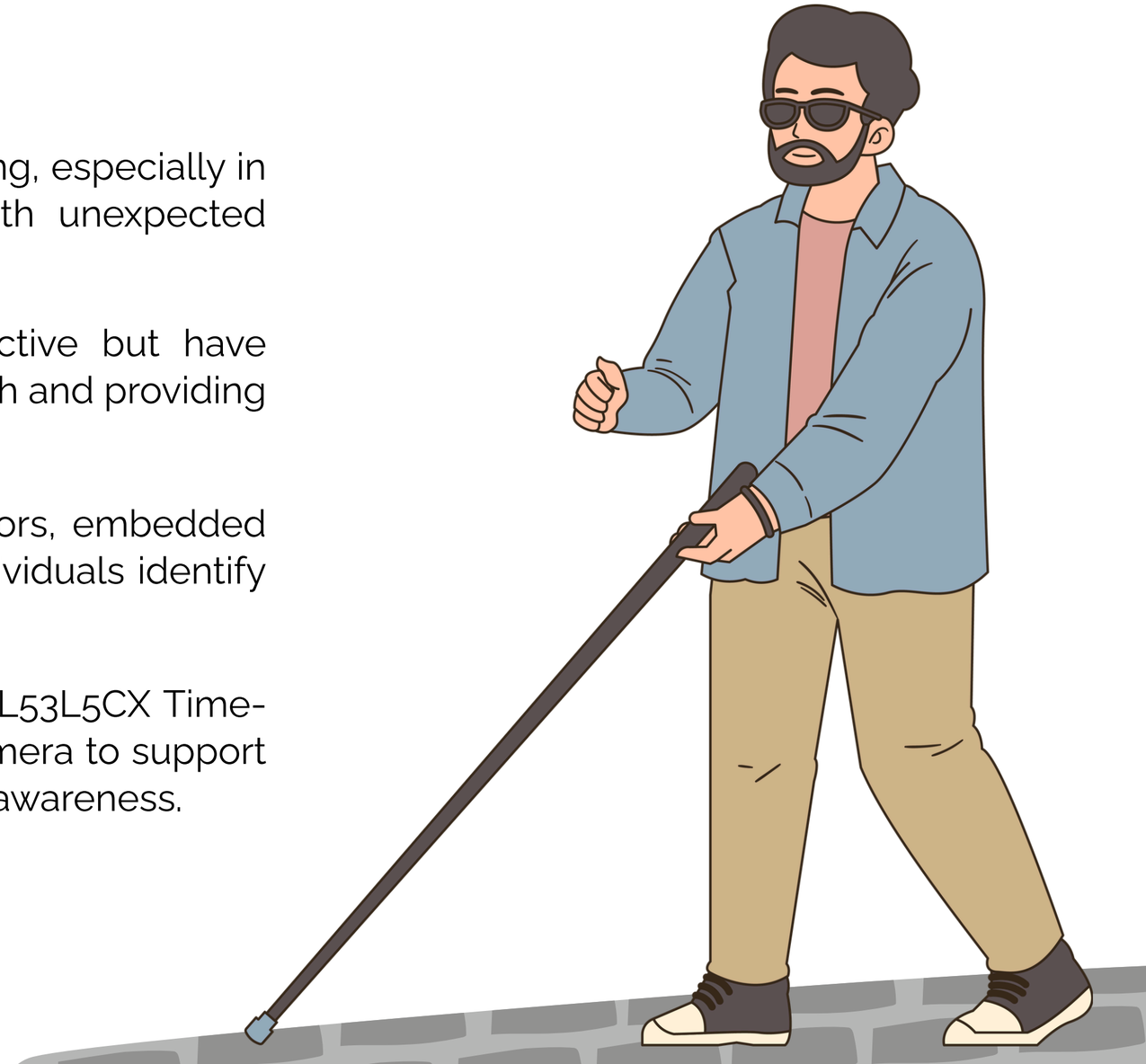
Overview:

Visual impairment can make everyday navigation challenging, especially in unfamiliar environments, crowded spaces, and areas with unexpected obstacles.

Traditional mobility aids, such as white canes, are effective but have limitations in detecting obstacles beyond their physical reach and providing contextual information.

An IoT-Based Assistive Navigation System integrates sensors, embedded systems, and intelligent alerts to help visually impaired individuals identify nearby obstacles and navigate more safely.

Our proposed system uses an ESP32-based controller, a VL53L5CX Time-of-Flight (ToF) distance sensor, an MPU6050 IMU, and a camera to support obstacle detection, movement analysis, and environmental awareness.



Introduction

Importance:

- independent mobility
- detect nearby obstacles
- provide real-time alerts.
- reduce dependence on others
- make navigation safer and more accessible

WHY IS THE NEED?



23.5%

Could not afford cataract surgery

Among people aged 50 and above who were blind due to cataracts, 23.5% reported affordability as a barrier to surgery.

19.1%

Affordability was a barrier even with less severe impairment

Among people with cataract-related visual impairment, 19.1% reported that surgery was unaffordable.

2.2B+

2.2 billion+

According to the WHO, at least 2.2 billion people worldwide have near or distance vision impairment, highlighting the need for accessible assistive technologies that support safer, more independent mobility

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Literature Review

	KEY CONTRIBUTION/ FINDINGS	RESEARCH GAP	INTERFACE OF DRISHTI
YOLO-OD: Obstacle Detection for Visually Impaired Navigation Assistance **Liping Yang and Chunliang Wang.** Publication year: 2024 | Journal: Sensors, 24(23), 7621 https://doi.org/10.3390/s24237621	Proposes YOLO-OD with a Feature Weighting Block, Adaptive Bottleneck Block, and Enhanced Feature Attention Head to improve detection of obstacles of different sizes, including small and partially occluded objects in outdoor scenes.	The model focuses on improving visual detection; this does not by itself establish complete navigation safety or end-to-end wearable performance.	Our project will investigate combining object classification with independent ToF distance measurements, motion information, and audio/vibration alerts. We must measure whether this fusion improves reliability rather than assume it does.
A Wearable Obstacle Avoidance Device for Visually Impaired Individuals with Cross-Modal Learning **Yun Gao, Dan Wu, Jie Song, Xueyi Zhang, Bangbang Hou, Hengfa Liu, Junqi Liao, and colleagues.** Publication year: 2025 | Journal: Nature Communications, 16, Article 2857 https://doi.org/10.1038/s41467-025-58085-x	Integrates video and depth sensing, smartphone-based cross-modal detection, and audio/haptic feedback. The authors report collision-avoidance rates of 100% in their evaluated scenarios, end-to-end latency below 320 ms, and around 11 hours of operation.	The system uses specialized hardware, including a custom FPGA-based processing unit, and is more complex than a basic student prototype	Our proposed distinction is to investigate a lower-cost ESP32-CAM and smartphone architecture with a separate ToF sensor and IMU. Cost, latency, reliability, and power consumption must be measured against the more advanced design.

	KEY CONTRIBUTION/ FINDINGS	RESEARCH GAP	INTERFACE OF DRISHTI
Towards a Real-Time Indoor Object Detection for Visually Impaired Users Using Raspberry Pi 4 and YOLOv11: A Feasibility Study **Ayman Noor, Hanan Almukhalfi, Arthur Souza, and Talal H. Noor** Publication year: 2025 | Journal article | DOI: 10.32604/cmes.2025.068393 https://doi.org/10.32604/cmes.2025.068393	Examines a wearable IoT architecture using Raspberry Pi 4 and YOLOv11 for indoor object detection, with wearable, network, cloud, and detection layers.	Raspberry Pi-based inference requires a separate computing board. In addition, in present time, it is costlier approach.	Our proposed architecture moves the primary AI inference workload to a smartphone, potentially reducing wearable hardware requirements. The comparison should examine inference latency, network dependence, cost, and battery life.
Enhancing Object Detection in Assistive Technology for the Visually Impaired: A DETR-Based Approach **Sunnia Ikram, Imran Sarwar Bajwa, Sujan Gyawali, Amna Ikram, and Najah Alsubaie.** Publication year: 2025 | Journal: IEEE Access, 13, 71647–71661 https://doi.org/10.1109/ACCESS.2025.3558370	Compares DETR, YOLOv8, and Faster R-CNN for assistive object detection and discusses optimization for mobile deployment. The reported results include 98% precision and a 40 ms processing time per frame for DETR.	High object-detection performance does not automatically establish accurate physical distance or safe navigation decisions.	Our system will investigate the relationship between model predictions, ToF measurements, and alert decisions. We should also benchmark YOLO inference on the actual target phone rather than compare model accuracy figures alone
An RGB-D Camera-Based Wearable Device for Visually Impaired People: Enhanced Navigation with Reduced Social Stigma Publication year: 2025 | Journal: Electronics, 14(11), 2168 Zhiwen Li ·, Fred Han · and Kangjie Zheng https://www.mdpi.com/2079-9292/14/11/2168	Combines RGB-D sensing, an IMU, point-cloud processing, and YOLO-based object recognition. It provides directional haptic feedback and audio prompts describing obstacles and their distances.	RGB-D hardware and point-cloud processing can add cost and computational requirements.	Our project will explore a more modular arrangement using an ESP32-CAM, VL53L5CX ToF sensor, and MPU6050, with detection performed on the phone. The trade-off is that a single-point ToF sensor does not provide the same depth information as an RGB-D camera.

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Literature Review

Research Gap

These are the gaps that emerge from comparing the papers

01

Visual recognition versus physical distance

Many approaches use camera-based detection or depth estimation. Our proposed design will evaluate camera-based object identification alongside direct ToF measurements, testing whether the two sources complement each other.

02

Processing cost and hardware complexity

Some advanced systems require Raspberry Pi boards, stereo cameras, RGB-D sensors, or specialized processors. Our proposed architecture will explore whether an ESP32-CAM and smartphone can provide a practical, lower-cost alternative.

03

Sensor integration and motion awareness

Camera and distance information can be affected by occlusion, alignment, and environmental conditions. Our project will investigate the additional value of IMU measurements and a common sensor-fusion decision process.

04

End-to-end validation

Detection accuracy alone does not establish that a wearable aid works reliably. Our evaluation should measure latency, distance error, false alerts, missed obstacles, power consumption, and performance under different conditions.

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Problem Definition

Problem Statement :

"Our project aims to address the limitations of **conventional mobility** aids by developing an **affordable** IoT-based **assistive navigation system** that uses sensors to detect nearby obstacles, analyze movement, and provide **timely alerts** to support safer and more **independent mobility** for visually impaired individuals."

Context

- Traditional mobility aids have limitations in detecting obstacles beyond their reach.
- Visually impaired people face challenges navigating crowded and unfamiliar environments.
- Affordable smart navigation solutions are still needed.



Significance

- Reduces the risk of collisions and injuries.
- Promotes independent and confident mobility.
- Improves accessibility through affordable assistive technology.

Objectives

Main Objective:

To develop an affordable IoT-based assistive navigation system that combines sensor fusion and YOLO-based object detection to support safer, more independent mobility for visually impaired individuals.



Specific Objectives

Obstacle Detection:

Use a VL ToF sensor to measure the distance to nearby obstacles.

Movement Analysis

Use the MPU6050 IMU to estimate device movement and orientation.

Object Recognition

Use a camera to capture images and a mobile phone to perform YOLO-based object detection over LAN Wi-Fi.

Real-Time Alerts

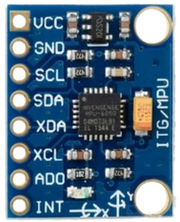
Provide timely audio or vibration alerts based on detected obstacles and recognized objects.

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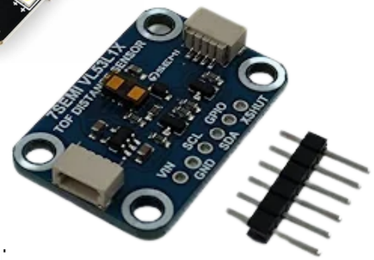
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Methodology









1

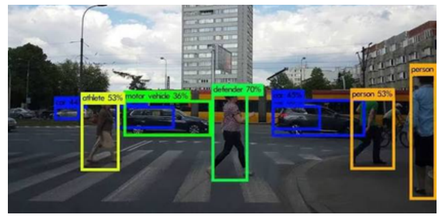
Data Acquisition

VL53L5CX and VL53L1X ToF measures obstacle distance; MPU6050 captures motion data; camera captures images

2

Edge Offloading

Camera images are sent to a mobile phone over LAN Wi-Fi for YOLO object detection.



3

Sensor Fusion

Distance, motion, and object detection results are combined when available.

4

Rule-Based Decision Making

Predefined distance thresholds determine the warning level; YOLO adds object information.

5

Alert Generation

Audio or vibration alerts inform the user about nearby obstacles



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Methodology

Data Acquisition



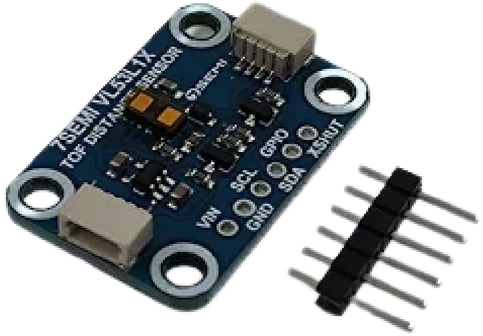
ESP32-CAM

Captures images of the surrounding environment for YOLO-based object detection and classification.



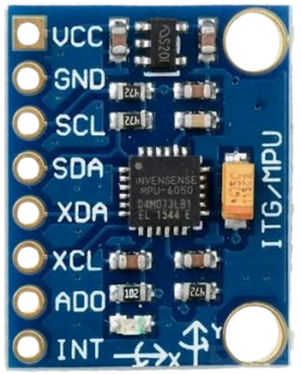
. VL53L5CX Multizone ToF Sensor

Forward view: Collects distance measurements of obstacles across 64 zones in its field of view.



VL53L1X

Downward view : Collects single-point distance measurements of the ground to identify potential changes in ground level.



MPU6050

Collects three-axis acceleration and three-axis angular velocity data to monitor the wearable's motion and orientation.

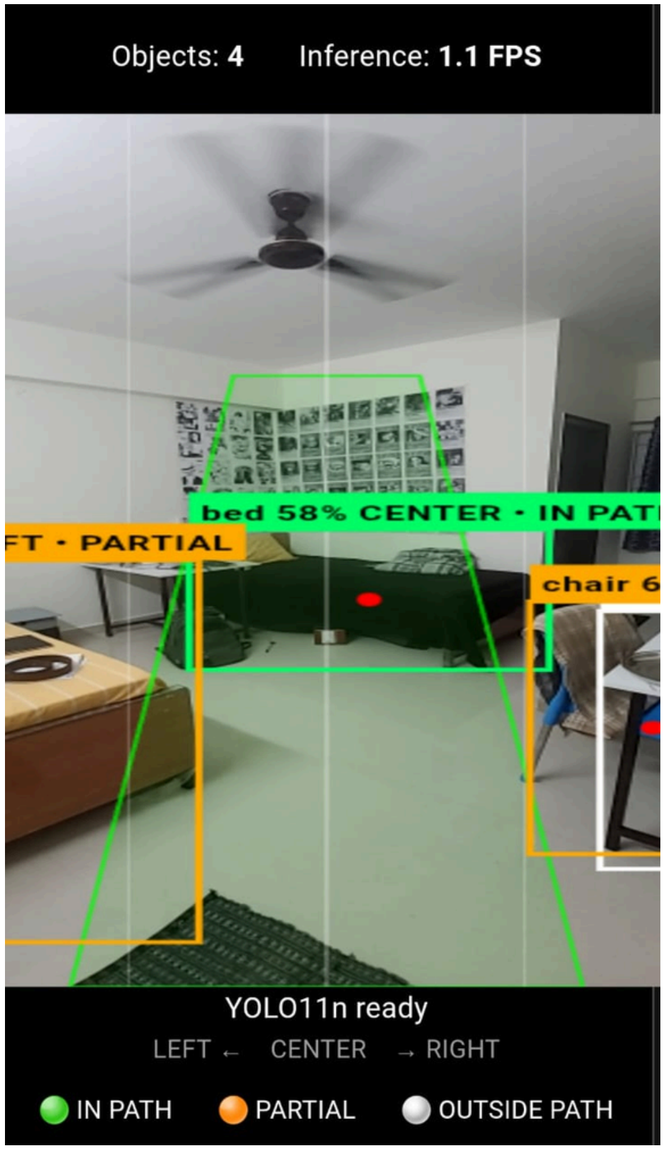
Methodology

Data analysis

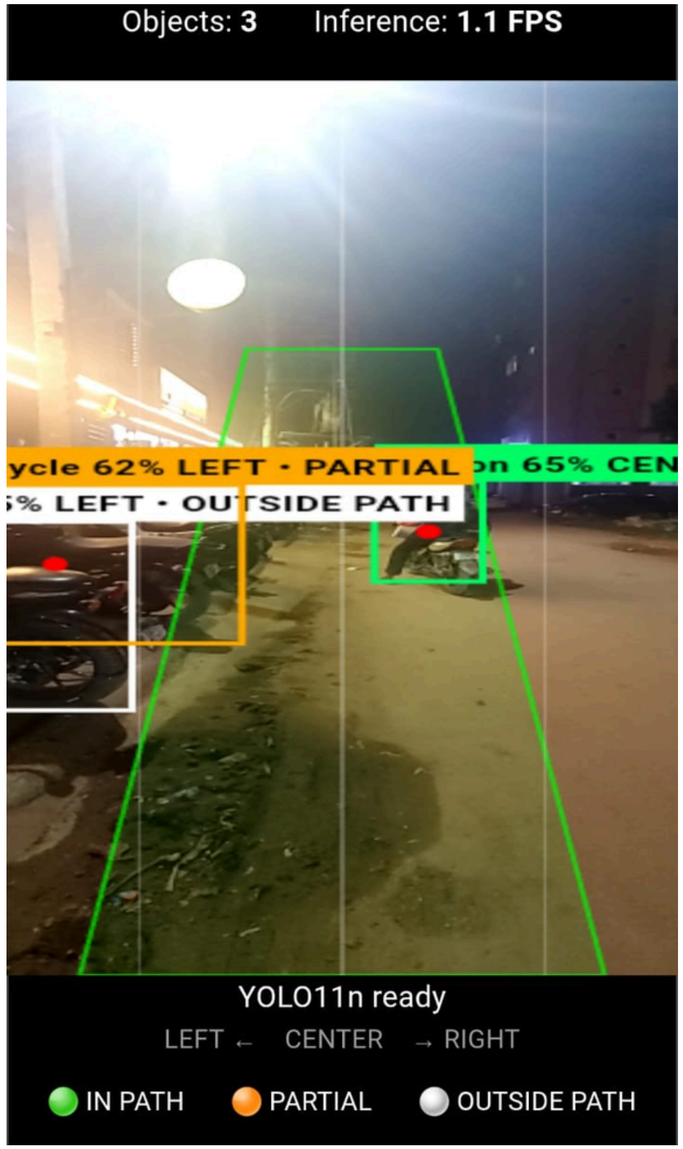
YOLO11n running locally on phone

Source Code:

https://github.com/bhavycan/BlindSight



Indoor



Outdoor

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Methodology

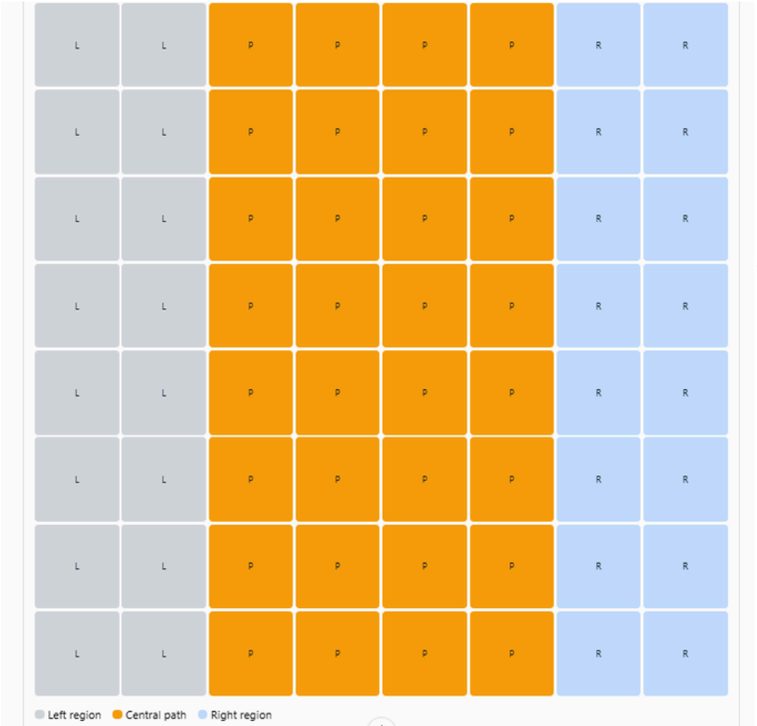
Data analysis

. VL53L5CX
Multizone ToF Sensor

Forward view: Collects distance measurements of obstacles across 64 zones in its field of view.

1800	1700	1600	1500	1500	1600	1700	1800
1600	1500	1400	1200	1200	1400	1500	1600
1500	1300	1000	700	700	1000	1300	1500
1400	1200	800	450	450	800	1200	1400
1400	1100	750	350	350	750	1100	1400
1500	1300	1000	650	650	1000	1300	1500
1600	1500	1400	1200	1200	1400	1500	1600
1800	1700	1600	1500	1500	1600	1700	1800

Very close Close Medium Farther



Distributing the 8 * 8 grid according to our path and calculating, is the obstacle in the path or not

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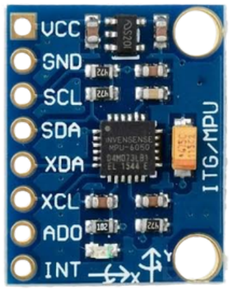
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Methodology

Data analysis

MPU6050

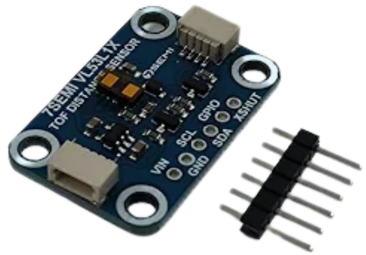
- Get the tiltness of the wearable device and adjust how the ToF zones are interpreted when the wearable tilts.
- Emergancy SOS (Future objective)



Increasing Accuracy

VL53L1X

- Downward motion object detection



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Methodology

OUTPUT

Vibration Motor

Will provide the different intensity of vibration according to the distance between person and obstacle.



distance > 4m

Clear
No Aleart

2 <= distance <= 4

Caution
Slow vibration pulse

1<= distance <2

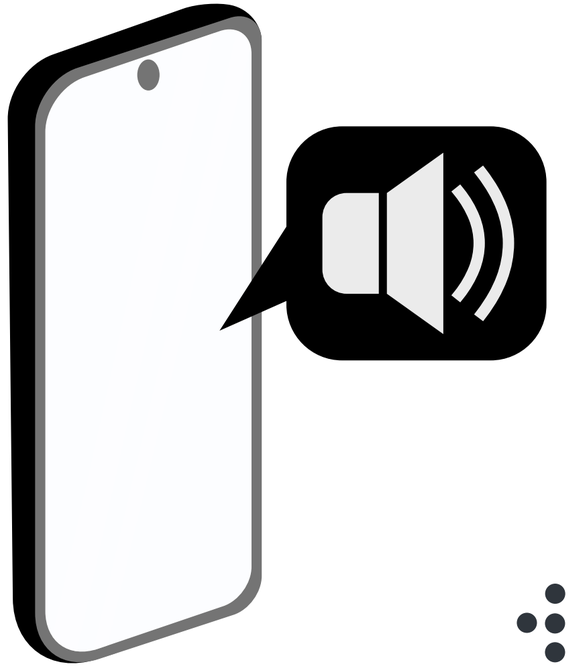
Warning
Fast pulse + audio name

distance < 1m

Danger
Continous

Phone Speaker

To describe the obstacle detected with the help of YOLO



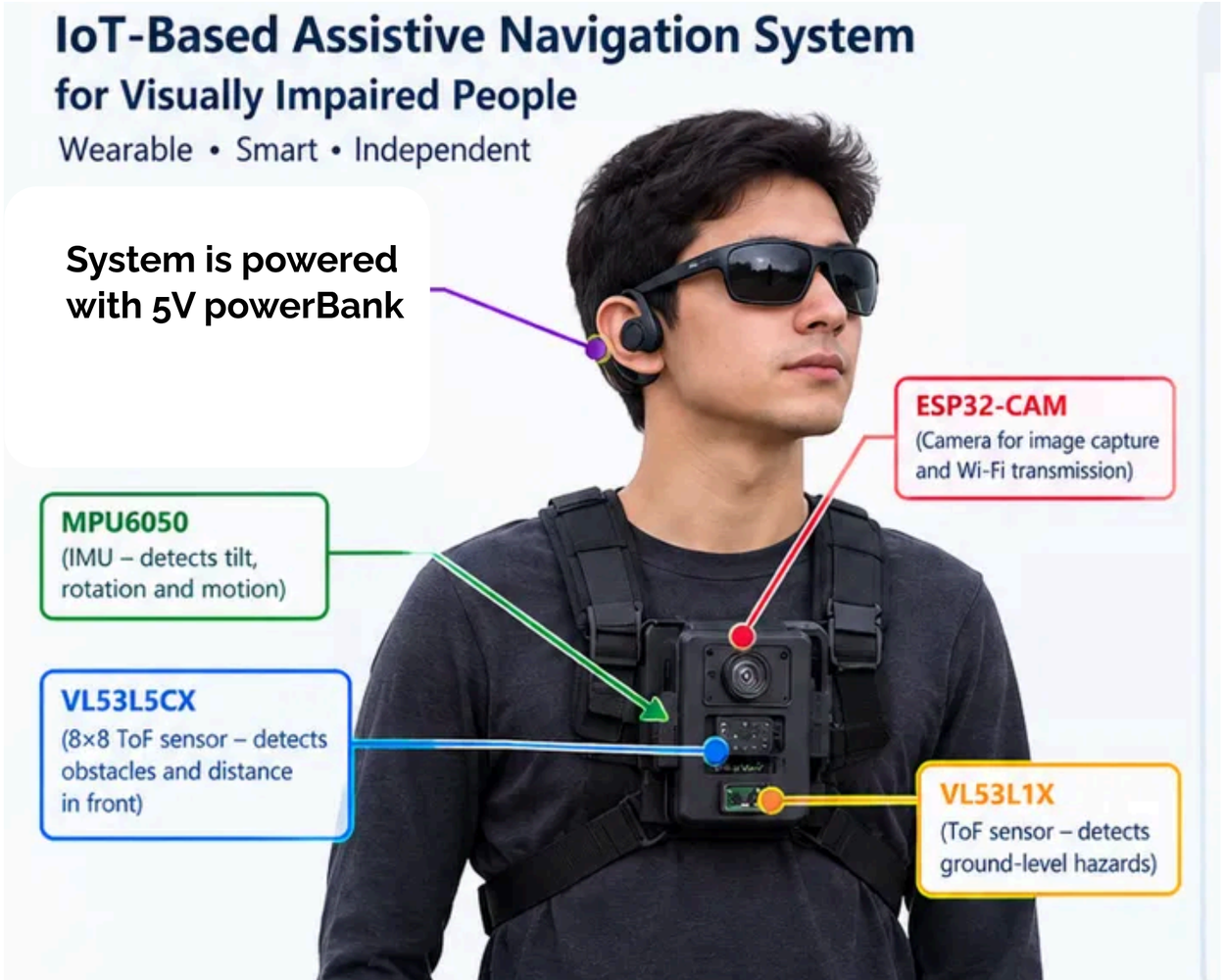
Methodology
illustration

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Power

System is powered with 5V powerBank

Image is generated using gpt, just for the illustration purpose



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Proposed Outcomes

What the project aims to deliver (targets, not results)

Expected Results

- Wearable prototype that identifies obstacles with YOLO
- Measures distance with VL53L1CX and MPU6050, VL53L1x
- Gives vibration and audio alerts

Potential Impact

- Phone-side detection keeps the wearable light
- Combines "what" and "how far" in one device
- Practical aid for independent mobility

Role modeling

- Outdoor and low-light testing
- Tuning alert thresholds with user feedback
- Extra sensors or route guidance

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Procedures: Timeline and Milestones

Planning

3-4 Week

Detailed plan; split modules among five members; budget

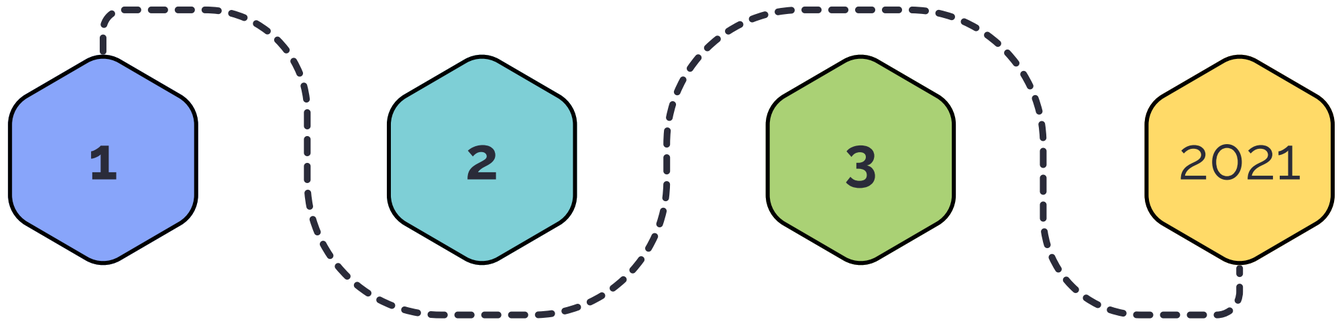
M2: design and test plan frozen

Closure

11-13 Week

Bench tests; analysis; report; panel review

M5: final review ready



M : Milestone

Initiation

1-2 Week

Define goals; list components; feasibility study

M1: scope and component list agreed

Execution

5-10 Week

Bring up ESP32-CAM and VL53L1X; YOLO on phone; decision logic; alerts; battery

M3 (wk 6): camera and sensor work alone. M4 (wk 10): integrated prototype

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Conclusion

What the proposed design combines

	MANY REPORTED SYSTEMS	OUR PROPOSED DESIGN
SENSING	Either camera recognition or range sensing	Camera (what) + VL53L1X (how far), VL53L1Cx
PROCESSING	Detection and alert logic on one unit	YOLO on a phone; ESP32 handles decisions
FEEDBACK	Single alert channel	Vibration and audio alerts
STATUS	Varies by study	Proposal stage; hardware not yet tested

Next Step:

Bring up the ESP32-CAM and VL53L1CX individually, then integrate the alert logic.

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References

YOLO-OD: Obstacle Detection for Visually Impaired Navigation Assistance
Liping Yang and Chunliang Wang.
Publication year: 2024 | Journal: Sensors, 24(23), 7621
https://doi.org/10.3390/s24237621

A Wearable Obstacle Avoidance Device for Visually Impaired Individuals with Cross-Modal Learning
Yun Gao, Dan Wu, Jie Song, Xueyi Zhang, Bangbang Hou, Hengfa Liu, Junqi Liao, and colleagues.
Publication year: 2025 | Journal: Nature Communications, 16, Article 2857
https://doi.org/10.1038/s41467-025-58085-x

Towards a Real-Time Indoor Object Detection for Visually Impaired Users Using Raspberry Pi 4 and YOLOv11: A Feasibility Study
Ayman Noor, Hanan Almukhalfi, Arthur Souza, and Talal H. Noor
Publication year: 2025 | Journal article | DOI: 10.32604/cmes.2025.068393
https://doi.org/10.32604/cmes.2025.068393

Enhancing Object Detection in Assistive Technology for the Visually Impaired: A DETR-Based Approach
Sunnia Ikram, Imran Sarwar Bajwa, Sujan Gyawali, Amna Ikram, and Najah Alsubaie.
Publication year: 2025 | Journal: IEEE Access, 13, 71647–71661
https://doi.org/10.1109/ACCESS.2025.3558370

An RGB-D Camera-Based Wearable Device for Visually Impaired People: Enhanced Navigation with Reduced Social Stigma
Publication year: 2025 | Journal: Electronics, 14(11), 2168
Zhiwen Li *, Fred Han * and Kangjie Zheng
https://www.mdpi.com/2079-9292/14/11/2168

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Expected Outcome by the End of First Year

01

If the prototype shows novel functionality: draft description and claims, run a prior-art search.

Patent Filing

02

Full paper or abstract in IEEE format presenting the prototype and key findings.

Conference Publication

03

Paper on methodology, prototype design, testing and results, for a Scopus-indexed journal.

Journal Publication

04

Extend into a larger project with objectives, methodology, impact and budget estimate.

Research Proposal

05

Potential Product to use in day to day life

Product

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